

Project Documentation

AI Fit Track API

Project Title

Fit Sense - Personalized fitness recommendations powered by AI.

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1. Introduction

AI FitTrack API is an intelligent fitness management system developed to help users manage and monitor their fitness activities. The system provides APIs for handling user information, workout records, fitness measurements, health-related details, and progress information.

The application uses a backend API to communicate between the user interface and the database. The API allows users to register, log in, view their fitness details, update their information, record workouts, and track their progress. All the data can be stored and managed securely in the database.

AI technology can be integrated into the system to analyze fitness information and provide personalized recommendations based on the user's activities, fitness level, and goals. It can help users understand their workout patterns and make better fitness decisions.

The system is designed with a simple and organized structure so that different applications can communicate with the same backend. It can be connected with web applications, mobile applications, and other fitness platforms.

AI FitTrack API can also manage different types of fitness data such as workout duration, exercise type, calories burned, weight, height, and other measurements. By maintaining these records, users can easily monitor their fitness progress over time.

The main purpose of AI FitTrack API is to make fitness tracking simple, organized, and accessible. The system reduces manual data management and provides a structured way to store and retrieve fitness information.

The project can be further enhanced by integrating wearable devices, real-time activity tracking, advanced AI-based recommendations, fitness goal prediction, and progress visualization. These features can make the system more useful for users and provide a better personalized fitness experience.

2. Abstract

AI FitTrack API is a fitness tracking system developed to help users manage and monitor their fitness information through a centralized platform. The system provides REST APIs for user registration, authentication, fitness data management, workout tracking, progress monitoring, and recommendation generation. It provides a simple and organized way to maintain fitness-related information.

The backend receives requests from the client application, processes the information, and stores the required data in the database. Users can create an account, log in securely, add their personal fitness details, record workout activities, and monitor their fitness progress over time. The system can manage information such as workout type, workout duration, fitness measurements, calories burned, fitness goals, and progress records.

The AI component can analyze the collected fitness information and generate suitable recommendations based on the user's activities and fitness goals. These recommendations can help users understand their workout patterns and maintain regular fitness activities. The system can also help users keep their fitness records organized and easily accessible.

The system follows a REST-based architecture, allowing the backend to communicate with different client applications. The API can be integrated with web applications, mobile applications, and other fitness platforms. The database provides organized storage and management of user and workout information.

AI FitTrack API is designed to be flexible and easy to extend. Future enhancements can include wearable device integration, real-time activity tracking, fitness progress charts, personalized workout plans, calorie analysis, and advanced AI-based recommendations.

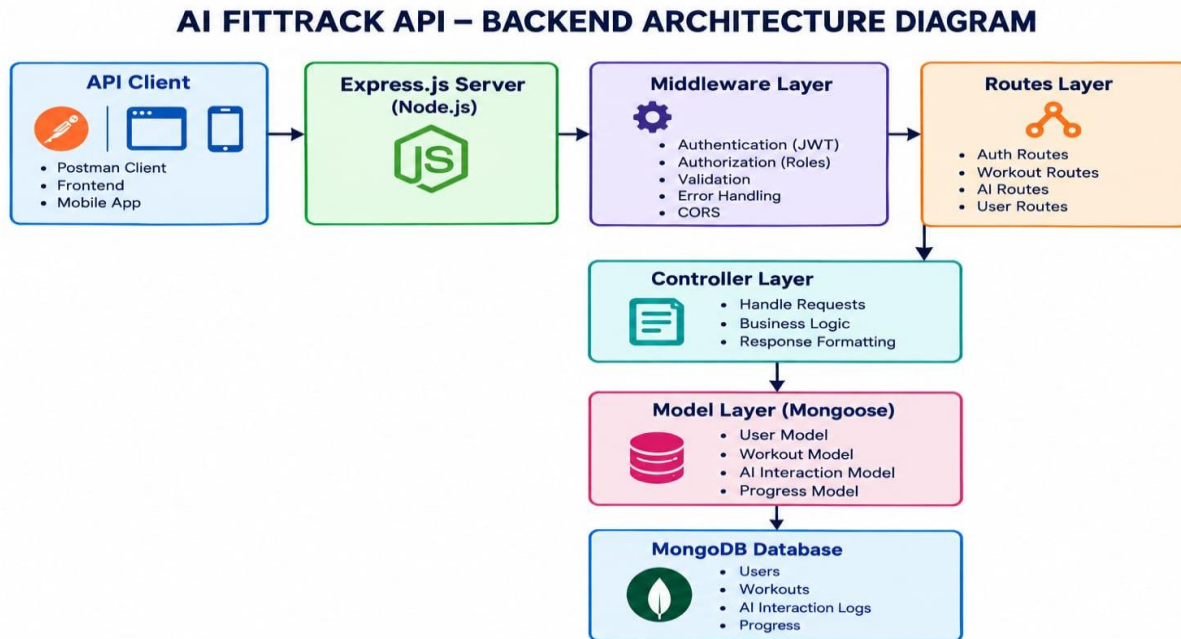
The main objective of AI FitTrack API is to provide a simple, organized, and intelligent platform for fitness management. By combining REST API technology, database management, and AI-based analysis, the system helps users manage their fitness information and monitor their progress effectively.

3.Objectives

The main objective of AI FitTrack API is to develop a simple, organized, and intelligent system for managing and monitoring fitness-related information. The system is designed to provide users with an easy way to maintain their fitness records and track their progress.

- **Centralized Fitness Management:** To provide a centralized platform for managing user profiles, fitness details, workout records, and recommendations.
- **User Management:** To allow users to register, log in, and manage their personal and fitness information.
- **Workout Tracking:** To record workout details such as workout type, duration, calories burned, and workout date.
- **Fitness Data Management:** To store and manage fitness measurements, goals, and other related information in an organized manner.
- **Progress Monitoring:** To help users monitor their fitness activities and track their progress over time.
- **AI-Based Recommendations:** To analyze available fitness information and generate suitable recommendations based on user activities and fitness goals.
- **REST API Development:** To develop REST APIs that enable communication between the frontend, backend, and database.
- **Database Management:** To use MySQL for storing and retrieving user, workout, fitness, and recommendation data.
- **CRUD Operations:** To provide Create, Read, Update, and Delete operations for effective management of fitness information.
- **Application Integration:** To provide an API structure that can be integrated with web and mobile applications.

4. Project Architecture



The FitTrack AI follows a layered architecture based on the MVC (Model–View–Controller) design pattern. The application separates routing, business logic, database interaction, middleware, and AI services into independent modules, making the project easier to maintain and scale.

When a client sends an HTTP request, the Express server receives it and forwards it to the corresponding route. Protected routes first pass through the authentication middleware, which validates the JWT token before granting access. Controllers process validated requests and invoke business logic or AI services as required. Database interactions are handled through Mongoose models, while Google Gemini AI is accessed through a dedicated service layer. Finally, the controller returns a structured JSON response to the client.

This layered approach improves modularity, security, and maintainability while ensuring that each component has a single responsibility.

5. Problem & Solution

Problem

Many users find it difficult to maintain and monitor their fitness records regularly. Workout information, body measurements, fitness goals, and progress data may be recorded manually or stored in different applications. This makes it difficult to maintain a complete and organized fitness history.

Users may also face difficulties in calculating and updating fitness information manually. The absence of a centralized platform can make it harder to access previous workout records and monitor fitness progress over time. In addition, general fitness suggestions may not always match the user's activities, fitness level, or goals. Managing fitness data across different applications can also create difficulties in accessing and sharing the required information.

Solution

AI FitTrack API provides a centralized solution for managing fitness information in an organized manner. The system allows users to create accounts, securely log in, add their fitness details, record workout activities, and monitor their progress over time. The database stores the required user and fitness information so that it can be easily retrieved and updated when needed.

The API handles communication between the frontend application and backend system. It provides a flexible way to manage user information, workout records, fitness measurements, and progress data. The AI component can analyze the available fitness information and generate suitable personalized recommendations based on the user's activities and fitness goals.

By combining REST APIs, database management, and AI-based analysis, AI FitTrack API reduces manual fitness record management and provides an organized platform for tracking fitness activities. The system can also be integrated with different web and mobile applications, making fitness information more accessible and easier to manage.

6. Software Requirements

The AI FitTrack API requires suitable software tools and technologies for development, testing, database management, and API communication.

Operating System: Windows 10 or Windows 11 can be used for developing and testing the application.

Programming Language: Python is used for backend development because it is simple and supports web development, database connectivity, and AI libraries.

Backend Framework: Flask or FastAPI is used to create RESTful APIs and handle requests between the frontend and backend.

Database: MySQL is used to store user details, workout records, fitness measurements, goals, progress, and recommendations.

Frontend: HTML, CSS, and JavaScript can be used to develop the user interface and communicate with the API.

API Testing Tool: Postman is used to test API endpoints, requests, responses, and error handling.

Development Environment: Visual Studio Code is used for writing, editing, testing, and managing the project source code.

Version Control: Git is used to track code changes and maintain different versions of the project.

Web Browser: Google Chrome or Microsoft Edge can be used to access and test the application.

Python Packages: Required Python packages are used for API development, database connectivity, data processing, and AI-based functionality. These software tools provide the necessary environment for developing, testing, and maintaining the AI FitTrack API system.

7. Hardware Requirements

The AI FitTrack API system does not require high-end hardware for basic development, testing, and usage. A standard computer with sufficient memory, storage, and internet connectivity is enough to run the application and related software.

Minimum Requirements

Processor: Intel Core i3 or equivalent processor.

RAM: Minimum 4 GB RAM is required to run the development tools and application.

Storage: At least 10 GB of free storage space is required for the project files, software, and database.

Monitor: A standard display is sufficient for development and testing.

Keyboard and Mouse: Standard keyboard and mouse are required for operating the system.

Internet: Internet connection is required for installing Python packages, accessing online services, and optional AI or cloud-based features.

Recommended Requirements

Processor: Intel Core i5 or equivalent processor is recommended for better performance.

RAM: 8 GB RAM or above is recommended for smooth development and testing.

Storage: 20 GB or above is recommended for storing project files, software, database records, and additional packages.

Internet: A stable broadband connection is recommended for API testing, package installation, and cloud-based services. These hardware requirements provide a suitable environment for developing, testing, and running the AI FitTrack API system efficiently.

8. System Architecture

AI FitTrack API follows a layered client-server architecture. The architecture separates the user interface, API handling, business logic, AI processing, and database operations. This structure makes the system organized, flexible, and easier to maintain.

Architecture Flow

User → Frontend → REST API → Controller → Business Logic → AI Layer → Database

The user interacts with the frontend to enter or view fitness information. The frontend sends requests to the REST API, which validates and processes the data.

The business logic performs the required operations, while the AI layer analyzes fitness data and generates suitable recommendations. The backend stores and retrieves data from the MySQL database and sends the results to the frontend in JSON format.

Main Components

Client Layer: Provides the user interface through which users interact with the system.

API Layer: Handles HTTP requests and responses and provides REST API endpoints.

Controller Layer: Receives requests, validates input, and connects the API with the required operations.

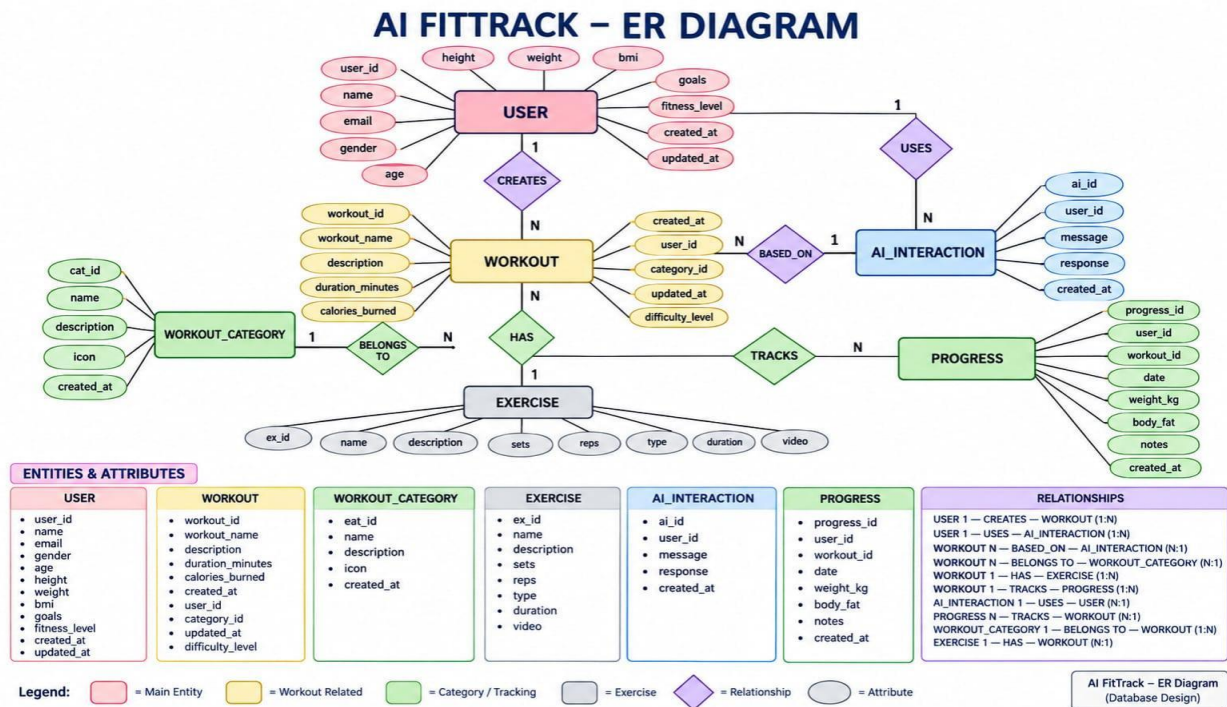
Business Logic Layer: Performs application operations such as managing users, workouts, and fitness records.

AI Layer: Analyzes fitness information and generates personalized recommendations.

Database Layer: Stores and retrieves user, workout, fitness, and progress data using MySQL.

This layered architecture helps separate different functions of the system and makes the AI FitTrack API easier to develop, test, maintain, and extend.

9. ER Diagram Explanation



The Entity Relationship Diagram (ER Diagram) represents the database structure of AI FitTrack. It shows the main entities, their attributes, and the relationships between them.

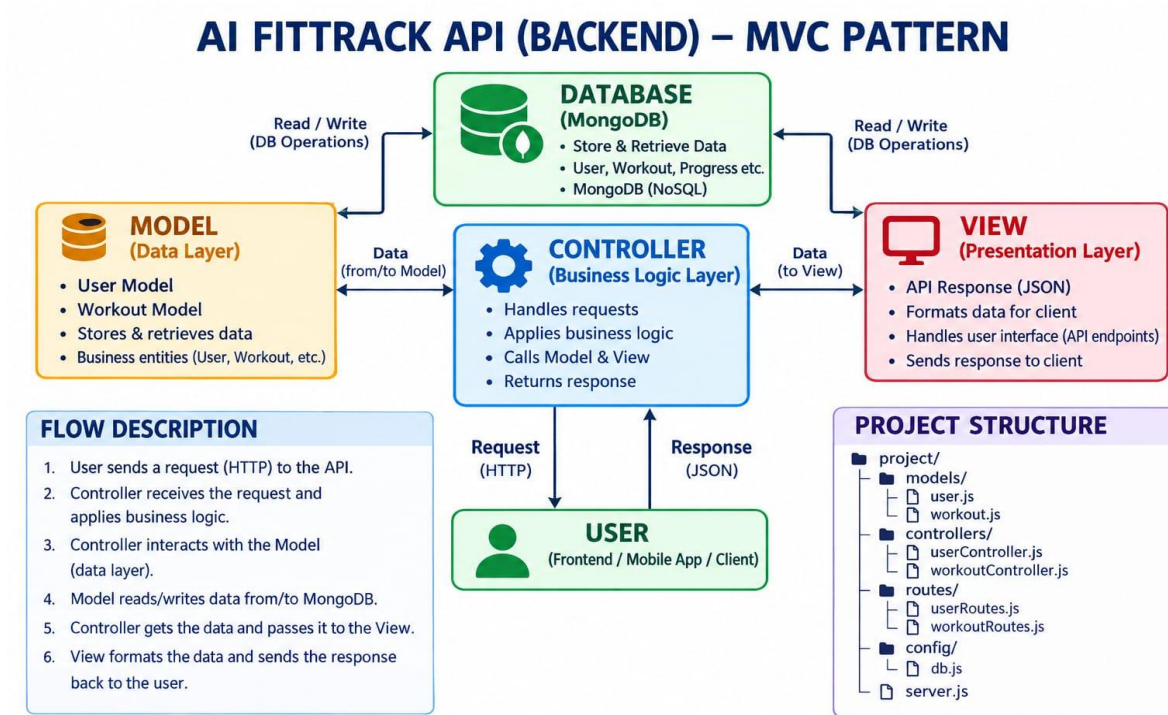
User Entity: Stores user details such as User ID, Name, Email, Password, Age, and Gender.

Workout Entity: Stores workout details such as Workout ID, User ID, Workout Type, Duration, Calories Burned, and Workout Date.

Fitness Data Entity: Stores fitness measurements such as Fitness ID, User ID, Height, Weight, BMI, and Fitness Goal.

Recommendation Entity: Stores AI-generated recommendations such as Recommendation ID, User ID, Recommendation Text, and Created Date.

10. MVC Architecture



MVC stands for **Model, View, and Controller**. It is a software architecture pattern that separates different parts of an application.

Model: Manages application data and database operations such as insert, update, delete, and retrieve.

View: Represents the user interface and displays fitness details, workout history, progress, and recommendations.

Controller: Receives user requests, processes the business logic, communicates with the Model, and sends the response.

MVC Flow:

User → View → Controller → Model → Database

This separation makes the application easier to develop, test, maintain, and update.

11. API Documentation

API stands for Application Programming Interface. It allows different software applications to communicate with each other. AI FitTrack uses REST APIs to exchange data between the frontend and backend.



User APIs

Method	Endpoint	Description
POST	/api/register	Register a new user
POST	/api/login	Log in a user
GET	/api/users/{id}	Get user details
PUT	/api/users/{id}	Update user details



Workout APIs

Method	Endpoint	Description
POST	/api/workouts	Add a new workout
GET	/api/workouts/{user_id}	View workout history
PUT	/api/workouts/{id}	Update workout details
DELETE	/api/workouts/{id}	Delete a workout



Fitness APIs

Method	Endpoint	Description
POST	/api/fitness	Add fitness data
GET	/api/fitness/{user_id}	View fitness data
PUT	/api/fitness/{id}	Update fitness data



Recommendation APIs

Method	Endpoint	Description
POST	/api/recommendations	Generate a fitness recommendation
GET	/api/recommendations/{user_id}	View user recommendations

Sample Request

```
{  
  "user_id": 101,  
  "workout_type": "Running",  
  "duration": 30,  
  "calories": 250  
}
```

12.Features

The main features of AI FitTrack API are as follows:

- **User Registration and Authentication:** Allows users to create an account and securely log in.
- **User Profile Management:** Allows users to add, view, and update their personal and fitness details.
- **Fitness Data Management:** Stores information such as height, weight, BMI, and fitness goals.
- **Workout Tracking:** Records workout type, duration, calories burned, and workout date.
- **Workout History:** Maintains previous workout records for reviewing activities.
- **Fitness Progress Monitoring:** Helps users track and monitor their fitness progress.
- **AI-Based Recommendations:** Provides suitable recommendations based on fitness activities and goals.
- **REST API Integration:** Enables communication between frontend and backend applications.
- **MySQL Database Integration:** Stores and manages user, workout, fitness, and recommendation data.
- **JSON-Based API Responses:** Uses JSON format to exchange data between applications.
- **CRUD Operations:** Supports Create, Read, Update, and Delete operations.
- **Web and Mobile Integration:** Allows the API to be integrated with web and mobile applications.

13. User Flow

The user flow explains how a user interacts with the AI FitTrack API system from registration to viewing fitness progress and recommendations.

Step 1 – Registration:

The user creates an account by entering the required personal details.

Step 2 – Login:

The registered user logs into the system using valid login credentials.

Step 3 – Profile:

The user enters or updates personal and fitness information such as age, gender, height, weight, and fitness goals.

Step 4 – Add Workout:

The user records workout details such as workout type, duration, calories burned, and workout date.

Step 5 – Store Data:

The API validates the entered information and stores it in the MySQL database.

Step 6 – Data Analysis:

The system processes the available fitness and workout information for analysis.

Step 7 – AI Recommendation:

The AI component analyzes the user's fitness data and can generate suitable personalized fitness recommendations.

Step 8 – Dashboard:

The user can view fitness information, workout history, progress, and AI-generated recommendations through the dashboard.

14. Advantages

- **Centralized Fitness Data:** User profiles, fitness details, workout records, and recommendations can be managed in one system.
- **Easy Workout Tracking:** Users can add, update, delete, and view their workout records.
- **Progress Monitoring:** Users can monitor their fitness information and progress over time.
- **Personalized Recommendations:** AI can analyze available fitness information and provide suitable suggestions.
- **REST API Support:** REST APIs allow easy communication between frontend and backend using HTTP and JSON.
- **Organized Database:** MySQL stores user, fitness, workout, and recommendation data in a structured manner.
- **Easy Integration:** The API can be integrated with web and mobile applications.
- **Modular Design:** Separate modules make user management, workout management, and AI features organized.
- **Easy Maintenance:** MVC and layered architecture make the system easier to develop and maintain.
- **Scalable System:** The API architecture provides a foundation for adding new features in the future.
- **CRUD Operations:** The system supports creating, reading, updating, and deleting fitness information.
- **Simple User Flow:** The system provides a clear flow from registration and login to fitness tracking, AI analysis, recommendations, and progress viewing.

15. Disadvantages

- **Internet Dependency:** Internet access may be required for installing packages and using optional AI or cloud services.
- **Manual Data Entry:** Users need to enter their fitness and workout information manually.
- **Data Accuracy:** AI recommendations depend on the quality and availability of the user's fitness data.
- **Limited Real-Time Tracking:** The current system does not provide complete real-time activity tracking.
- **No Wearable Integration:** The current version does not directly connect with smartwatches or other wearable devices.
- **Basic AI Functionality:** The current system provides a basic foundation for AI recommendations and can be improved with advanced AI models.
- **Mobile Application Not Included:** The current project is mainly API-based and does not include a dedicated mobile application.
- **Limited Analytics:** The system provides basic progress monitoring and can be extended with detailed fitness analytics.
- **Regular Updates Required:** Users need to regularly update their workout and fitness information to maintain useful records.
- **System Expansion Required:** Additional features and integrations are needed to support advanced fitness-management requirements.

16. Conclusion

AI FitTrack API provides a structured and efficient platform for managing fitness-related information. It allows users to maintain their profiles, record workout activities, store fitness data, and monitor their progress through an API-based system.

The system uses REST API technology to provide communication between the frontend and backend. Python is used for backend development, while MySQL is used to store and manage user, workout, fitness, and recommendation data. This combination provides an organized and flexible approach to fitness data management.

The AI component provides a foundation for analyzing available fitness information and generating suitable recommendations based on user activities and fitness goals. The system also supports important operations such as user management, workout tracking, progress monitoring, and CRUD operations.

The layered and modular architecture makes the system easier to develop, maintain, and extend. The API-based design also allows the system to be integrated with different web and mobile applications.

In the future, AI FitTrack API can be enhanced with wearable-device integration, dedicated mobile applications, real-time activity tracking, advanced AI models, personalized workout plans, and detailed fitness analytics. These enhancements can improve the functionality and usefulness of the system.

Overall, AI FitTrack API provides a strong foundation for developing a simple, organized, and intelligent fitness management system. It combines API technology, database management, and AI functionality to support effective fitness tracking a